an API needs to be considered as a whole, and not as individual operations. You'll learn more about HSMs in chapter 9.

- Last, there may be security vulnerabilities due to the implementation of the API. For example, failing to check the size of inputs to your API may allow an attacker to bring down your server by sending a very large input that consumes all available memory; a type of *denial of service* (DoS) attack.

  DEFINITION A ***denial of service*** (or DoS) attack occurs when an attacker can prevent legitimate users from accessing a service. This is often done by flooding a service with network traffic, preventing it from servicing legitimate requests, but can also be achieved by disconnecting network connections or exploiting bugs to crash the server.

Some API designs are more amenable to secure implementation than others, and there are tools and techniques that can help to ensure a secure implementation. It is much easier (and cheaper) to think about secure development before you begin coding rather than waiting until security defects are identified later in development or in production. Retrospectively altering a design and development lifecycle to account for security is possible, but rarely easy. This book will teach you practical techniques for securing APIs, but if you want a more thorough grounding in how to design-in security from the start, then I highly recommend the book *Secure by Design* (Manning, 2018).